

Fiazul Haque

AI & Automation Engineer

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PROFESSIONAL SUMMARY

AI & Automation Engineer with ~2 years of experience designing production-grade intelligent systems. At Rokomari.com, built end-to-end AI pipelines - from natural-language SQL interfaces and SEO automation to distributed image-generation platforms on Google Cloud Vertex AI. Strong command of fault-tolerant async architectures, LLM integration, and scraping infrastructure that runs unsupervised for days.

WORK EXPERIENCE

AI Engineer — Rokomari.com

Feb 2025 - Present

Dhaka, Bangladesh

- Built a natural-language-to-SQL interface over a complex legacy database using LLaMA 3, eliminating ~80-90% of ad-hoc data team requests.
- Architected an end-to-end product image generation platform on Google Cloud Vertex AI (Nano banana) with internal API, human verification layer, batch JSONL workflows, and automated GCS upload — documented in full README with Docker Compose setup.
- Designed a content generation & SEO automation system (Gemini Pro/Flash + FastEmbed semantic search) that auto-generates product descriptions, maps messy specs to canonical schemas, and publishes directly to the CMS — with Redis-backed idempotency locks to prevent duplicate AI API calls.
- Engineered a bulk image background-removal pipeline using BriaAI RMBG v2.0 with edge refinement; handled ~90% of catalog images reliably at scale.
- Built hybrid Selenium + Playwright scrapers with CAPTCHA detection and handling, IP rotation, and multi-day autonomous operation — zero human supervision.
- Developed an SEO validation system (NLTK + spaCy) automating grammar, entity, and keyword checks; reduced content review time significantly.

Python Developer — UXToDo

Nov 2023 - Dec 2024

Dhaka, Bangladesh

- Built AI proof-of-concepts including a medical chatbot and semantic search tool using OpenAI and local LLMs.
- Integrated LLM APIs into production-ready client-facing services with unit and integration test coverage.
- Contributed to a confidential enterprise project ensuring system reliability through structured testing workflows.

NOTABLE PROJECTS

GCP Product Image Generator

- Distributed image generation system on Vertex AI with Celery workers, Redis broker, PostgreSQL, FastAPI, and a dedicated image-serving container - all orchestrated via Docker Compose.
- Sequential pipeline (every 2 min) handles prompt generation, GCS upload, JSONL batching, Vertex AI submission, and result processing with guardrails to ensure that each token has business impact - fully automated.

Content Generation & SEO Automation

- Semantic specification mapper using FastEmbed vector embeddings + cosine similarity with regex numerical guardrails to prevent false matches (e.g., 75mAh vs 7500mAh).
- Race-condition-safe background pipeline: DB-level UNIQUE constraints + SQLAlchemy IntegrityError rollbacks + dynamic Redis lock expiry - zero duplicate AI calls in production.

NLP-to-SQL AI Interface

- Natural language query interface over a legacy MySQL database using LLaMA 3 with chain-of-thought prompting - deployed internally at Rokomari.com.

Autonomous Google Search Automation Suite

- Production-grade scraper with human-behavior simulation, runs multi site in parallel with a round robin, IP rotation, session recovery, and progress tracking. Runs for days without supervision.

Demand Intelligence & Opportunity Gap Detection

- Multi-signal scoring engine (Google Ads API, GA4 clickstream, scraped competitor catalogs, distributor availability) with a ReAct agent that autonomously investigates gaps and drafts full procurement category plans - semantic matching layer handles 100k+ monthly queries with zero manual tagging.
- Production-grade scraper with human-behavior simulation, CAPTCHA detection, IP rotation, session recovery, and progress tracking. Runs for days without supervision.

TECHNICAL SKILLS

Languages	Python (primary), JavaScript, SQL, Bash
AI / ML	LLM integration & prompting (LLaMA 3, Gemini, OpenAI), agentic ReAct loops, RAG pipelines, Semantic search & Embeddings, Vector database, NLP, audio and image processing
Backend	FastAPI, Django, Celery, async pipelines, REST APIs
Cloud	Google Cloud Platform (Vertex AI, GCS, Nano banana), Docker
Automation	Selenium, Playwright, stealth scraping
Databases	PostgreSQL, MySQL, SQLite, SQLAlchemy ORM, Redis
Integrations	Facebook Graph API, GCP Vertex AI, Rokomari internal API

EDUCATION

B.Sc. in Computer Science & Engineering

Bangladesh University of Business and Technology (BUBT) **CGPA: 3.73 / 4.00**

Feb 2019 - Dec 2023